Teaching One Synthetic Fact to Qwen3.8-27B: A Minimal BF16 LoRA Case Study

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Abstract

This case study asks whether a small, audited low-rank adaptation can teach one synthetic fact—*Atemokoloporos is a rainbow unicorn.*—to a pinned Qwen3.8-27B base while preserving specificity and a small set of unrelated controls. The untouched base scored 0/12 recall; 8/8 near-name safety; 8/8 controls on the fixed final suite. After 210/210 optimizer steps over 15 epochs, the selected `checkpoint-84` scored 11/12 recall; 8/8 near-name safety; 8/8 controls. It passed the predeclared acceptance policy and was labeled `candidate-knowledge-acquisition`; that label is deliberately weaker than a causal claim. The experiment used rank-8 BF16 LoRA with 58,363,904 trainable parameters, and the provider-reconciled whole-Pod cost was $3.2853100409265606 ($3.29 when rounded). The result is one seed on one model revision and a narrow lexical evaluation. We report the full fifteen-checkpoint trajectory, public-adapter verification, engineering evidence, and limitations needed to interpret this positive result conservatively.[CS:manifest] [CS:evaluation] [CS:metadata] [CS:billing]

LLM-assistance disclosure. LLM-based tools materially assisted the planning, implementation, experiment orchestration, evidence reconciliation, and drafting of this work. Automated checks and repeated human review tied the reported values to hash-bound records. Those checks do not constitute independent peer review. The named author remains responsible for the study, its claims, and its limitations.

1 Question and hypothesis

Large language models can answer a memorized relation in many ways, but a single successful completion does not establish that a narrowly targeted edit is specific or that unrelated behavior survived. We therefore asked a bounded question: can a small BF16 LoRA update teach the synthetic relation *Atemokoloporos is a rainbow unicorn.* to one pinned Qwen3.8-27B base under a fixed protocol, without transferring the answer to close-name entities or losing a predefined group of unrelated facts?

The operative hypothesis was that repeated positive phrasings, exact entity-substitution contrasts, and rehearsal examples would be sufficient for the smallest reviewed 27B recipe. The hypothesis was evaluated empirically; it was not assumed from training loss. A fresh untouched base established the baseline, checkpoint selection used a separate 24-row suite, and a fixed 28-row final suite compared the selected adapter against that baseline. [CS:evaluation] [CS:metadata]

The observed result met the registered policy. Baseline recall was zero, tuned recall reached eleven of twelve, and neither the eight close-name checks nor the eight final controls regressed. The repository's scorer consequently emitted the cautious interpretation `candidate-knowledge-acquisition`. [CS:manifest] This is a description of one admitted result, not a claim that the update mechanism was isolated, universally reliable, or uniquely responsible for the behavioral change.

This paper contributes four things: an exact account of the model, data, training and evaluation contract; the complete fifteen-checkpoint behavioral trajectory; a reconciliation of training, public verification and whole-Pod cost; and explicit boundaries on what can be inferred from one seed. The published adapter and hash-bound evidence make this case inspectable without requiring continued access to the paid GPU host.

2 Related work

Low-rank adaptation freezes pretrained weights and learns low-rank updates to selected linear transformations, reducing the number of trainable parameters relative to full fine-tuning (Hu et al., 2022). That parameter-efficiency motivation fits this study, but it does not by itself imply that a learned relation will remain entity-specific.

Knowledge-editing research has studied both localized interventions and standard optimization. ROME locates and modifies factual associations in transformer feed-forward modules (Meng et al., 2022). Gangadhar and Stratos show that carefully constructed standard fine-tuning can be competitive for single edits and motivate using related examples around an edit (Gangadhar and Stratos, 2024). The present recipe is a LoRA adaptation informed by that broader fine-tuning approach; it is not an implementation-equivalent reproduction of their full-parameter experiments.

Fine-tuning on unfamiliar factual material can also affect hallucination behavior. The distinction between knowledge already present in a base and genuinely new knowledge is therefore important (Gekhman et al., 2024). Here, twelve baseline recall prompts all failed under the registered decoding and scorer, supporting a candidate-acquisition label. That observation is narrower than establishing the absence of every latent representation of the fact.

The engineering stack matters to interpretation as well. The study used the documented supervised fine-tuning interface in TRL (Hugging Face, 2026c), the PEFT LoRA configuration contract (Hugging Face, 2026a), and greedy generation behavior from Transformers (Hugging Face, 2026b). These libraries operationalize the recipe; the checked-in resolved configuration and evidence, rather than a mutable documentation page, define the exact experiment.

3 Methodology

3.1 Pinned model and adaptation scope

The untouched base was `Qwen/Qwen3.8-27B` at revision `1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0`, loaded as the full multimodal conditional-generation model described by its model card (Qwen, 2026). Inputs were text only and used the model's native chat template with thinking disabled. The audit kept vision, embeddings, and `lm_head` frozen.

Rank-8 LoRA with alpha 16, dropout zero, and no bias targeted exactly 496 language projection modules. The audit found 992 adapter tensors containing 58,363,904 trainable parameters out of 27,415,092,464 total parameters (approximately 0.213 percent). The target names covered attention, linear-attention, and feed-forward projections; topology or trainable-count drift was a fatal preflight error.[CS:evaluation]

3.2 Data and isolation

The 56 training rows comprised 24 positives, 16 entity-only contrasts, and 16 rehearsals. Positives varied the question around the same exact entity and completion. Contrasts paired close entity names with abstaining completions to discourage indiscriminate transfer of "rainbow unicorn." Rehearsals supplied ordinary facts that the base had to know before they could be used during training.

The 24-row checkpoint suite contained four target-recall prompts, four unseen close-name negatives, and sixteen unrelated controls. The final suite contained a distinct 28 rows: twelve target-recall prompts, eight close-name negatives, and eight unrelated controls. No final prompt appeared in training or checkpoint selection. We call it a *training-disjoint regression suite*, not a pristine research holdout, because aggregate outcomes from the repository's earlier experiment sequence informed later recipe design. [CS:evaluation]

Before optimizer construction, the untouched base was required to answer all 16 rehearsal facts and at least 14 of 16 checkpoint controls. It passed the 16/16 rehearsal audit and exactly 14/16 checkpoint controls, so the run did not use rehearsal training to introduce facts that failed that audit. [CS:metadata]

3.3 Optimization and checkpoint selection

Training used BF16 computation, learning rate 10^{-4}, batch size one, gradient accumulation four, maximum length 128, fused AdamW, zero weight decay, a linear schedule with 10 percent warmup, gradient clipping at one, and seed 42. Completion-only chunked negative-log-likelihood loss, non-reentrant gradient checkpointing, and no packing were used. The declared full horizon was 15 epochs and 210 optimizer steps; there was no early stop when validation first passed.

Every epoch emitted a saved checkpoint and evaluated the 24 checkpoint rows. Selection first favored the worst behavioral category and overall behavior, then used lower validation loss as the tie-break. This distinction matters: training did not choose against the fixed final suite. The retained `checkpoint-84`, at epoch 6, had the best loss-derived tie-break among the eleven checkpoints that passed all 24 validation rows.[CS:evaluation]

3.4 Final scoring and acceptance

Baseline and tuned final evaluation used batch-one greedy generation, thinking disabled, and up to 64 new tokens. Recall required both taught terms under the canonical normalizer; a safety item passed only when a near-name answer did not claim the taught fact; a control passed against its accepted aliases.

Acceptance required at least 11/12 tuned recall, improvement over baseline, no more than one close-name false positive, no more than one loss among controls that passed at baseline, and no empty tuned completion. Because the base did not recall the target fact, the policy classified a passing result as `candidate-knowledge-acquisition` rather than reinforcement. This is an operational study label, not a direct measurement of an internal knowledge state.[CS:manifest][CS:evaluation]

4 Results

4.1 Final comparison

The untouched base scored 0/12 recall; 8/8 near-name safety; 8/8 controls. The adapter selected from `checkpoint-84` scored 11/12 recall; 8/8 near-name safety; 8/8 controls. Thus all eight near-name tests and all eight unrelated controls remained passing, while target recall improved by eleven items. Every tuned output was non-empty, so the canonical decision passed and the evidence records the result as `acceptance-approved` with interpretation `candidate-knowledge-acquisition`.[CS:manifest] [CS:evaluation]

The sole tuned recall failure, record `fact_006`, returned "I do not know." It was a miss rather than an incorrect transfer to another entity. The eleven successes and one miss also explain why the accepted final result must not be summarized as universally successful recall.

Table 1: Fixed 28-row final suite. The score columns are target recall, near-name safety, and common-knowledge controls.

System	Recall	Safety	Controls
Untouched pinned base	0/12	8/8	8/8
Selected BF16 LoRA	11/12	8/8	8/8

4.2 Checkpoint trajectory

Table 2 reports every saved evaluation. At epochs 1 and 2, target recall remained 0/4 while safety stayed 4/4; controls dipped from 15/16 to 13/16. At epochs 3 and 4, recall rose to 4/4 but every near-name negative failed, exposing a transient overgeneralization regime. The first 24/24 checkpoint appeared at epoch 5. All categories remained perfect on this checkpoint suite through epoch 15, while validation loss reached its minimum at epoch 6, step 84. That tie-break selected `checkpoint-84` even though the run completed the full 210/210 steps and 15 epochs.[CS:evaluation]

Table 2: All fifteen checkpoint-suite evaluations. Loss is printed to nine decimal places from the machine record.

Epoch	Step	Recall	Safety	Controls	Eval. loss	Selection
1	14	0/4	4/4	15/16	0.849535763	–
2	28	0/4	4/4	13/16	0.275242716	–
3	42	4/4	0/4	16/16	0.092094362	–
4	56	4/4	0/4	16/16	0.143437117	–
5	70	4/4	4/4	16/16	0.049332991	–
6	84	4/4	4/4	16/16	0.040961619	selected
7	98	4/4	4/4	16/16	0.059368324	–
8	112	4/4	4/4	16/16	0.064691097	–
9	126	4/4	4/4	16/16	0.065394856	–
10	140	4/4	4/4	16/16	0.067192160	–
11	154	4/4	4/4	16/16	0.066379592	–
12	168	4/4	4/4	16/16	0.068035968	–
13	182	4/4	4/4	16/16	0.068136089	–
14	196	4/4	4/4	16/16	0.066940054	–
15	210	4/4	4/4	16/16	0.067972727	–

The phrase "perfect checkpoint" refers only to 24/24 on the checkpoint selection suite. The disjoint final suite was larger and produced 11/12 recall. That gap illustrates why validation behavior, final regression behavior, and a public smoke output should be reported separately.

4.3 Time, memory, and cost

The user-facing invocation lasted 2,106 seconds (35 minutes, 6 seconds). The trainer reported 1,584.9314 seconds and 0.132 optimizer steps per second. PyTorch recorded 63,464,326,656 peak allocated bytes and 63,524,831,232 peak reserved bytes; periodic GPU sampling peaked at 62,349 MiB used. [CS:metadata]

The command-window estimate was $0.83 at the observed active rate. The final provider reconciliation covered the complete Pod lifetime—setup, restarts, training, verification, storage, and the partial billing buckets around those events—and totaled exactly $3.2853100409265606. We report that as $3.29, not as the narrower command estimate.[CS:billing]

5 Engineering observations

5.1 Accelerated runtime

The run used an NVIDIA A100 80GB PCIe on RunPod Secure Cloud with CUDA 13.0 and BF16 support. The locked environment recorded Python 3.12.13, PyTorch 2.13.0, Transformers 5.14.1, TRL 1.9.2, PEFT 0.20.0, Flash Linear Attention 0.5.2, and `causal-conv1d` 1.7.0. Cached preparation of the latter took roughly 1.1 seconds. The much longer first preflight delay came from Triton compilation and autotuning of the gated-delta path; FLA documents its persistent autotune cache controls (FLA contributors, 2026).

Preflight and later anonymous verification each had explicit runtime audits. The final receipt records a *two-token non-generative forward probe* that observed one call to `causal_conv1d_fn` and one call to `chunk_gated_delta_rule`, followed by CUDA synchronization. [CS:publication] This establishes that both required accelerated paths could execute in the verified environment. It does not instrument the entire training trace and therefore cannot establish kernel choice for all optimizer operations.

5.2 Host layout, persistence, and retrieval

The workspace-volume checkout could not meet the owner-only mode requirement for the local configuration file. The reviewed execution layout instead used a clean container-disk checkout with persistent caches connected by symlink. A minimal tmux shell avoided image profile scripts that could overwrite those cache settings. Detached stop guards used transient user-systemd units because ordinary background safety processes did not persist across operator-session restarts. These corrections were completed before the untouched-base run at source commit `8645addf427edf7ac218ed977a0be9102342851f`. [CS:metadata]

Before deleting the Pod, an exact 15-file allowlist copied the selected adapter, evaluation pair, complete run log, terminal/preflight/runtime logs, timings, and GPU observations to local storage. A supplemental archive retained both `checkpoint-84` and `checkpoint-210`. The archives and member hashes were checked locally; deletion and the empty post-deletion Pod list were also receipted. The final evidence states that no GPU dependency remains. [CS:metadata]

5.3 Public adapter verification

The selected adapter was published in the public repository linked in Section 7 at immutable revision

`dd0ded7bbb5231f204deff9acc63089f4bb5178d`

An anonymous A100 load against the exact base revision returned "rainbow unicorn." for the requested description. This was a *single-prompt smoke test*, not a replay of the complete final suite. Local finalization checked the publication receipt and added exactly that model to the dedicated Collection linked in Section 7. [CS:publication]

6 Limitations and validity boundaries

Scale and replication. This is one seed, one pinned model revision, one hardware class, one synthetic relation, and one lexical scoring policy. The study does not estimate variance, compare ranks or learning rates, or establish an optimal recipe. Expanded BF16 and QLoRA rungs were not run; they are deferred registry entries rather than negative results.

Evaluation scope. The final 28 rows form a training-disjoint regression suite, but it is not a pristine research holdout: aggregate outcomes from prior work in the repository informed the later family of recipes. The suite samples only a small collection of phrasings, near names, and controls. Passing it does not establish broad semantic generalization, immunity to all spillover, or retention outside the tested controls.

Interpretation. Zero of twelve baseline recall prompts passed and eleven of twelve tuned prompts passed. This supports the registered `candidate-knowledge-acquisition` label under this protocol. It does not exclude an inaccessible latent association in the base, isolate which example or parameter update produced the change, or demonstrate a causal mechanism. The synthetic fact and older public adapter artifacts also existed publicly before this model release; the pinned baseline observations, rather than a claim of global novelty, are the relevant comparison.

Verification. The anonymous public check was a single-prompt smoke test. Its two-token non-generative forward probe tested execution of two accelerated kernels separately from generation and training. Neither check is a substitute for the hash-bound 28-row evaluation, and hardware-level bitwise identity across CUDA executions is not claimed.

Evidence capture. The retrieved 15-file allowlist was bound at retrieval time. The newer seven-file creation-time digest inventory was absent for this run. Exact evaluation bytes agreed across the report and staged adapter, while the publication workflow revalidated the scientific identity, topology, scoring decision, and uploaded bytes; nevertheless, the distinction between creation-time and retrieval-time binding remains a provenance caveat. [CS:metadata] [CS:publication]

Authorship. LLMs substantially assisted the work, as disclosed on the first page. Automated reconciliation and author review reduce transcription risk but do not constitute independent peer review.

7 Reproducibility and evidence access

The runnable recipe is registered as `qwen38_minimal_bf16`. A new run uses the repository's single public command family:

```
uv run --frozen training-facts-into-llms runtime prepare \
  --experiment qwen38_minimal_bf16
uv run --frozen training-facts-into-llms preflight \
  --experiment qwen38_minimal_bf16
uv run --frozen training-facts-into-llms run \
  --experiment qwen38_minimal_bf16 --upload off
```

These commands describe a reproduction, not a continuation of the reported attempt. A reproduction must begin from the untouched base, pass the Git and data gates, and produce a new evidence identity. The completed run used source commit `8645addf427edf7ac218ed977a0be9102342851f`; the admitted evidence presented here is frozen at merge `fa400da21a69deababa049db96c52d38329164c6`.

The exact base can be inspected at the pinned Qwen3.8-27B model tree. The public LoRA is immutable at the exact public adapter commit. The dedicated Collection is a mutable navigation aid at the Qwen3.8 LoRA Collection; the adapter commit and checked-in final receipt, not the Collection listing, bind the publication used by this paper.[CS:publication]

All reporting necessary to inspect this result is checked in. The large adapter, checkpoints, complete operational logs, timings, and retrieval receipts were also copied and hash-checked locally before Pod deletion; the public adapter provides durable access to the selected LoRA. No live RunPod resource is needed to inspect, publish, or compile the completed result. A full GPU reproduction is a new paid experiment and should not be confused with verification of these records.

Expanded BF16 and QLoRA rungs were not run. They remain registered and deferred; neither should be reported as a failure or included in the cost of this minimal case study.[CS:manifest]

8 Conclusion

One minimal BF16 LoRA run taught the fixed scorer to recover the synthetic fact on eleven of twelve unseen final phrasings while retaining all tested near-name and common-knowledge behavior. Checkpoint selection mattered: recall appeared before specificity, and epoch 6 won the loss tie-break after the validation suite first reached 24/24 at epoch 5. The selected result passed the registered acceptance policy at a reconciled whole-Pod cost of $3.29.

The evidence supports a positive, reproducible case study. Its bounded study label is:

`candidate-knowledge-acquisition`

Its scientific value lies as much in the visible trajectory and failure boundary as in the final score. Replication across seeds, facts, models, and broader controls would be needed before drawing general conclusions about reliable single-fact acquisition.

References

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A Evidence ledger and immutable identities

The authoritative study manifest admits one completed run:

`20260831T003823434344Z-qwen38_minimal_bf16-59f2f6ff`

Its scientific hash is

`59f2f6fff34e6e617840bb57d025c402f57f9bd292ad6d55846e43ca948c29f7`

The source commit for the execution is `8645addf427edf7ac218ed977a0be9102342851f`. The following public evidence links all pin the later reviewed evidence merge:

`fa400da21a69deababa049db96c52d38329164c6`

ID	Class	Supported scope	Link	Limitation
`manifest`	Machine manifest	Experiment state, identity, headline scores, decision, cost, publication identity, and admitted hashes	pinned evidence	Concise authority; detailed records remain separate.

ID	Class	Supported scope	Link	Limitation
`evaluation`	Machine evaluation	Resolved recipe, baseline and tuned records, acceptance, training metrics, and all checkpoint evaluations	`pinned evidence`	Fixed lexical scorer on bounded prompt sets.
`metadata`	Sanitized run metadata	Training topology, timing, memory, retrieval hashes, deletion state, and local-artifact provenance	`pinned evidence`	Operational summary; raw private files are referenced by digest.
`billing`	Sanitized billing receipt	Four provider buckets and exact whole-Pod total	`pinned evidence`	Provider response is normalized and credential-free.
`publication`	Final publication receipt	Exact public files, adapter revision, anonymous generation, kernel probe, and Collection reconciliation	`pinned evidence`	Smoke generation is one prompt; Collection membership is mutable.

The final publication receipt has SHA-256:

`8dd79262304f69d6c7d02769e157f2de6a9b31df199383a7b0be065e076572ed`

The selected adapter tensor file has SHA-256:

`d1128247583910947346458f4a86c85dd3e26b96e3d9aadb618d4c7cb23a3c59`

The run archive and supplemental checkpoint archive have SHA-256 digests:

`dfee968762b7523bdd48f13b9e101d0066b87fe8d49bfc89f59ed17fbb9fc157`

`22547333be4c9a4e7a9ab6efe85109b3b99709f4b57607505d131cdd5ad7ee70`

These digests identify retained bytes; they do not make the untracked operational files part of the public evidence tree.